

3.3.2 Gas exchange

Content	Opportunities for skills development
Adaptations of gas exchange surfaces, shown by gas exchange: • across the body surface of a single-celled organism • in the tracheal system of an insect (tracheae, tracheoles and spiracles) • across the gills of fish (gill lamellae and filaments including the counter-current principle) • by the leaves of dicotyledonous plants (mesophyll and stomata). Structural and functional compromises between the opposing needs for efficient gas exchange and the limitation of water loss shown by terrestrial insects and xerophytic plants. The gross structure of the human gas exchange system limited to the alveoli, bronchioles, bronchi, trachea and lungs. The essential features of the alveolar epithelium as a surface over which gas exchange takes place. Ventilation and the exchange of gases in the lungs. The mechanism of breathing to include the role of the diaphragm and the antagonistic interaction between the external and internal intercostal muscles in bringing about pressure changes in the thoracic cavity. Students should be able to: • interpret information relating to the effects of lung disease on gas exchange and/or ventilation • interpret data relating to the effects of pollution and smoking on the incidence of lung disease • analyse and interpret data associated with specific risk factors and the incidence of lung disease • evaluate the way in which experimental data led to statutory restrictions on the sources of risk factors • recognise correlations and causal relationships.	AT j Students could dissect mammalian lungs, the gas exchange system of a bony fish or of an insect. AT d Students could use an optical microscope to: • examine prepared mounts of gas exchange surfaces of a mammal, fish and insect, or temporary mounts of gills • examine vertical sections through a dicotyledonous leaf. AT b Students could use three-way taps, manometers and simple respirometers to measure volumes of air involved in gas exchange. MS 2.2 Students could be given values of pulmonary ventilation rate (PVR) and one other measure, requiring them to change the subject of the equation: $PVR = \text{tidal volume} \times \text{breathing rate}$